1 R Setup, renv, and Docker

1.1 R installation

Install the latest R release from CRAN. This book was developed with R 4.4.x. We recommend RStudio or VS Code with the R extension as your IDE.

1.2 Package management with renv

This project uses **renv** for reproducible package management. After cloning the repository:

```r
install.packages("renv")
renv::restore()
```

This installs the exact package versions recorded in `renv.lock`. To update the lockfile after adding a new package:

```r
renv::install("newpackage")
renv::snapshot()
```

1.3 Core packages

The table below lists the primary packages used throughout the book.

Package	Purpose	Chapter(s)
openalexR	OpenAlex API access	5–37
tidyverse	Data wrangling and visualisation	All
igraph	Network analysis	15–20
tidygraph	Tidy network manipulation	15–20
ggraph	Network visualisation	15–20
quanteda	Text mining and corpus management	21–25
topicmodels	LDA topic modelling	22, CS-5
stm	Structural topic models	22
gt	Publication-quality tables	Throughout
targets	Pipeline orchestration	34
plotly	Interactive charts	35
DT	Interactive data tables	35
shiny	Web applications	37
word2vec	Word embeddings	24
uwot	UMAP dimensionality reduction	24

1.4 Docker environment

A `Dockerfile` is provided at the repository root. It builds on `rocker/verse:4.4.1` and installs all required system libraries and R packages.

```bash
docker build -t scientometrics-in-r .
docker run --rm -p 8787:8787 -e DISABLE_AUTH=true scientometrics-in-r
```

Then open `http://localhost:8787` in your browser to access RStudio Server with all dependencies pre-installed.

1.5 Quarto

Install Quarto CLI version 1.4 or later. To render the book locally:

```
quarto render
```

Rendered output appears in `_book/`. For iterative development, use `quarto preview` to get live-reloading in the browser.